

5.3 The responses to climate change

This chapter will explain:

- ★ the strategies used to manage the impacts of climate change
- ★ how successful mitigation and adaptation strategies and techniques have been in managing the impacts of climate change
- ★ detailed specific example: the impacts of, and responses to, climate change in Bangladesh.

The strategies used to manage climate change – national and international agreements

The major international agreements to manage climate change include the Paris Agreement, the Kyoto Protocol and the United Nations Framework Convention of Climate Change (UNFCCC).

The **UNFCCC** is the United Nations' (UN) process to negotiate an agreement to limit global climate change. To achieve this, it proposes to limit the amount of greenhouse gases in the atmosphere. It was signed in 1992 at the UN Conference on the Environment (the 'Earth Summit') in Rio de Janeiro. Its main objective is to stabilise greenhouse gases in the atmosphere. By 2022, the UNFCCC had 198 members ('Parties'), who meet annually at COP (Conference of the Parties) meetings.

The **Kyoto Protocol**, signed in 1997, became effective in 2005. In 2020 there were 192 member countries. The Kyoto Protocol aims to reduce global climate change by decreasing atmospheric greenhouse concentrations to a level that would prevent dangerous human interference. The Protocol had differentiated responsibilities, recognising that HICs were historically responsible for most greenhouse gas emissions. The Protocol's first commitment period was 2008–12. Although many HICs reduced their emissions, total global emissions increased from 1990 to 2010. A second agreement existed from 2012 to 2020. Some 37 countries had binding targets, including the EU (28 countries at that time), Australia and Ukraine. Some countries withdrew from the Protocol, including Canada and Russia.

The **Paris Agreement** (2015) is an international treaty on climate change, agreed by 196 countries. It is a separate agreement under the UNFCCC. The long-term goal of the Paris Agreement is to keep global climate change to less than 2°C above pre-industrial levels, and ideally to less than 1.5°C above pre-industrial levels. To achieve this, greenhouse gas emissions need to be reduced and net zero (that is, when emissions of greenhouse gases are balanced by removal of greenhouse gases from the atmosphere) reached by about 2050. The Paris Agreement has been criticised by some for not being strict enough. In 2017, the USA, under President Trump, announced that it would pull out of the Paris Agreement. It withdrew in 2020, but rejoined in 2021 under President Biden. The Paris Agreement does not cover emissions from international aviation or shipping.

Despite these agreements, greenhouse gases in the atmosphere continue to rise, as do global temperatures.

An evaluation of mitigation and adaptation strategies

Climate change **adaptation** refers to methods of learning to live with climate change, whereas **mitigation** refers to measures to prevent or limit global climate change ([Figure 5.12](#)). Adaptation is easier but less effective. Mitigation would be better in the long term, but is more difficult to achieve.

Adaptation includes:

- » building higher flood walls
- » using more air conditioning systems to cool living/work spaces
- » ensuring availability of vaccines for diseases new to a region, such as malaria
- » building flood shelters
- » growing crops that are adapted to drought or heat.

Mitigation refers to attempts to reduce global climate change, including:

- » burning less fossil fuel
- » cutting down fewer trees
- » planting more vegetation
- » using sustainable transport
- » conserving energy
- » using renewable energy sources
- » eating less red meat and dairy products.

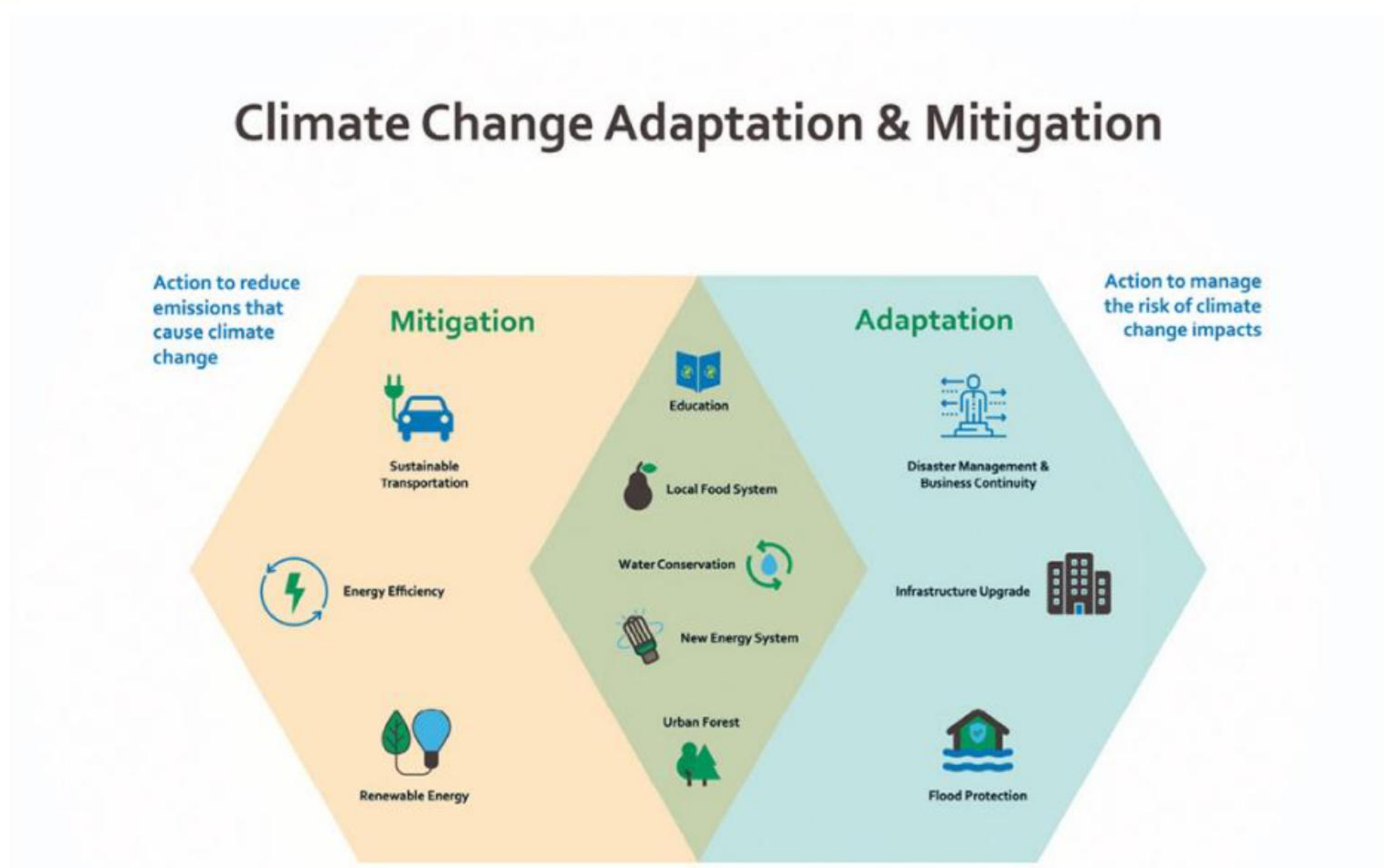
The disadvantages of mitigation include cost, for example the cost of developing nuclear power or building HEP schemes, and the difficulty of encouraging changes in people's behaviour, such as using less energy, using public transport, cycling or walking to work, college or school. The less mitigation that society achieves today, the more it will need to adapt in the future.

Adaptation may be easier, but it does not get rid of the problem. Adaptation will become progressively less effective if there is no mitigation, as global climate change will become more extreme over time. However, using clean energy sources, such as solar and wind power, will reduce the use of fossil fuels.

Other measures to reduce global climate change include insulating homes, switching off electricity sources when not in use, changing from a meat- and dairy-based diet to a vegetarian diet, reducing consumption of goods and using fewer plastics.

Activities

- 1 Suggest reasons why international agreements to manage the impact of climate change have been unsuccessful.
- 2 Explain how green infrastructure and water- and energy-conservation can be examples of adaptation and mitigation.



▲ **Figure 5.12** Adaptation and mitigation strategies for climate change

Detailed specific example

The impacts of, and responses to, climate change in Bangladesh

The impacts of climate change

Bangladesh accounts for less than 1 per cent of global greenhouse gas emissions, but it is one of the most vulnerable countries to climate devastation. This is known as a climate injustice.

Between 2000 and 2019 Bangladesh experienced economic loss of US\$3.7 billion and 185 extreme weather events. Around 90 million Bangladeshis, over 50 per cent of the population, have a ‘high’ exposure to flooding. Two-thirds of Bangladesh is less than 5 m above sea level ([Figure 5.13](#)) and around one-third of the population live in coastal areas. A 50 cm rise in sea level could cause Bangladesh to lose 11 per cent of its land and up to 18 million people would be forced to leave their homes. A sea level rise of 2 m could displace up to 50 million people. Rising sea levels also cause salinisation, which is when salt accumulates in soil and severely reduces the soil’s fertility.



▲ **Figure 5.13** Flood risk in Bangladesh

Extreme events are becoming more frequent in Bangladesh. Up to 50 per cent of people living in Bangladesh's informal settlements are believed to be there due to flooding elsewhere. Dhaka, the capital, now has a population density of up to 47,500 people/km².

Tropical cyclones are becoming more intense. For example, in 2007 Cyclone Sidr brought winds of 240 km/h and killed over 3400 people. In 2009, Cyclone Aila led to the deaths of 200 people and left about 200,000 people homeless. Cyclone Amphan in 2020 was the strongest cyclone in Bangladesh's history. It led to the deaths of 128 people and caused US\$14 billion worth of damage in Bangladesh and India.

The responses to climate change

To deal with such events, Bangladesh has developed a Climate Change Strategy Action Plan (BCCSAP). This was first produced in 2009, with five adaptation plans and one for mitigation. The Nationally Determined Contributions (NDCs) of 2021 planned:

- to reduce greenhouse gas emissions by >27 million tonnes of carbon dioxide equivalent (Mt CO₂e) (6.75 per cent) using Bangladesh's own resources by the end of 2030
- to reduce greenhouse gas emissions by >60 Mt CO₂e (>15 per cent) using international support by the end of 2030
- to reduce greenhouse gas emissions by around 90 Mt CO₂e (21.5 per cent) using combined resources by the end of 2030.

Strategies and techniques to manage climate change

Adaptation projects have included the building of embankments, over 4500 cyclone shelters and 300 flood shelters, and the excavation/re-excavation of canals.

Mitigation schemes have included the planting of millions of trees, including mangrove forests. As a result, emissions are expected to have been reduced by over 2 Mt CO₂e by 2025.

Bangladesh is pursuing a low carbon future with greater emphasis on renewable energy and increased energy efficiency. Farming methods used to adapt to the changing climate include:

- early-harvest, rapid-developing varieties, for example rice that develops in 100–20 days (traditional varieties require 140–50 days)
- drought-resistant early varieties
- salt-tolerant rice varieties
- flooding-tolerant crop varieties
- stress-tolerant crop varieties.

Bangladesh joined the Kyoto Protocol in 2013 and signed the Paris Agreement in 2016.

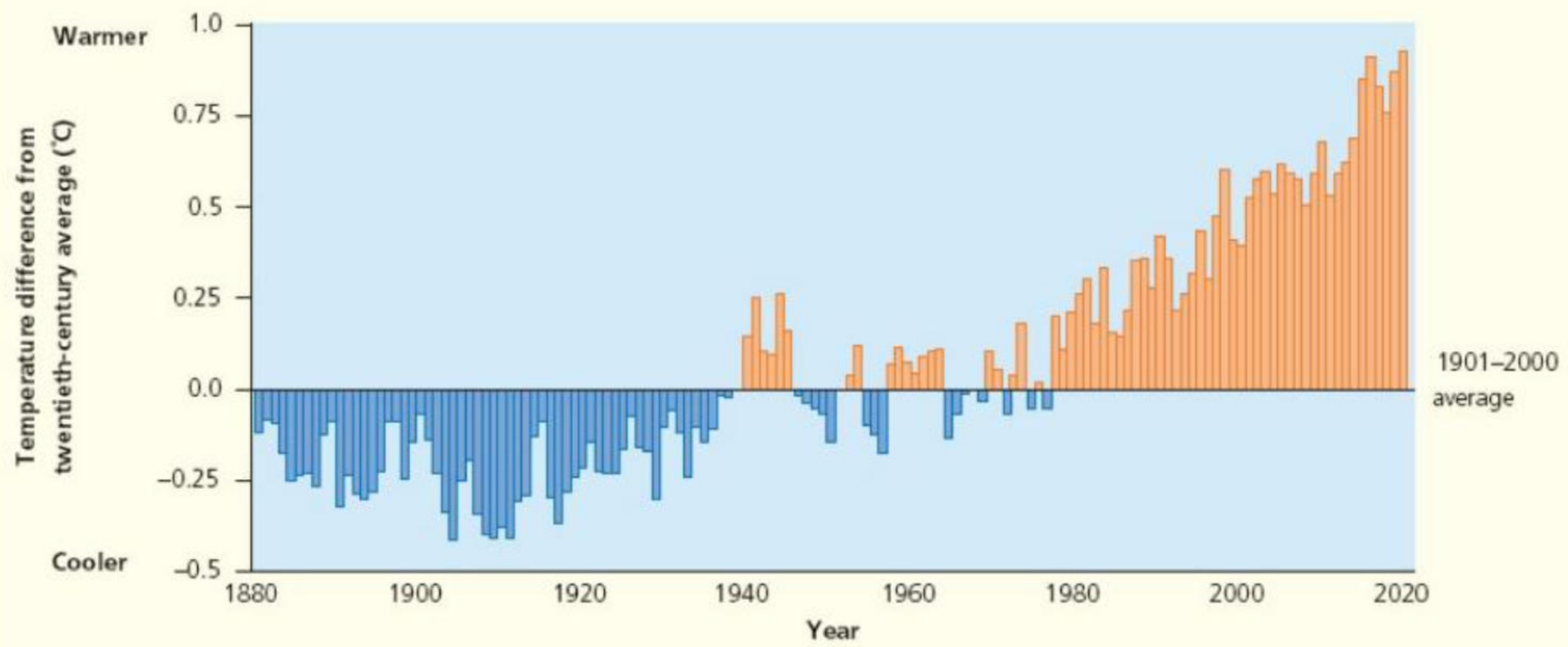
Activities

- 1 Describe the distribution and type of flood risk in Bangladesh, as shown in [Figure 5.13 \(page 120\)](#).
- 2 Distinguish between adaptation and mitigation strategies that have been used in Bangladesh.

Practice questions

- 1 [Figure 5.14](#) shows annual global surface temperature change for land and ocean, compared to the 1901–2000 average.
 - a Estimate the largest negative temperature difference from the twentieth-century average. [1]
 - b Estimate the largest positive temperature difference from the twentieth-century average. [1]
 - c Identify the decades when temperatures were lowest. [1]

- d** Identify the decade when temperatures were highest. [1]
- e** Estimate the increase in temperature between the decades with the lowest temperatures and the decade with the highest temperatures. [2]
- 2** Distinguish between the greenhouse effect and global warming. [4]
- 3 a** Define climate change mitigation and climate change adaptation. [2]
- b** Evaluate techniques of climate change mitigation and adaptation. [7]



▲ **Figure 5.14** Annual global surface temperature change for land and ocean, compared to the 1901-2000 average